

17-1c The Equilibrium for an Oligopoly

Oligopolists would like to form cartels and earn monopoly profits, but that is often impossible. Squabbling among cartel members over how to divide the profit in the market can make agreement among members difficult. In addition, antitrust laws prohibit explicit agreements among oligopolists as a matter of public policy. Even talking about pricing and production restrictions with competitors can be a criminal offense. Let's therefore consider what happens if Jack and Jill decide separately how much water to produce.

At first, one might expect Jack and Jill to reach the monopoly outcome on their own because this outcome maximizes their joint profit. In the absence of a binding agreement, however, the monopoly outcome is unlikely. To see why, imagine that Jack expects Jill to produce only 30 gallons (half of the monopoly quantity). Jack would reason as follows:

"I could produce 30 gallons as well. In this case, a total of 60 gallons of water would be sold at a price of \$60 a gallon. My profit would be \$1,800 (30 gallons $\times$ \$60 a gallon). Alternatively, I could produce 40 gallons. In this case, a total of 70 gallons of water would be sold at a price of \$50 a gallon. My profit would be \$2,000 (40 gallons $\times$ \$50 a gallon). Even though total profit in the market would fall, my profit would be higher because I would have a larger share of the market."

Of course, Jill might reason the same way. If so, Jack and Jill would each bring 40 gallons to town. Total sales would be 80 gallons, and the price would fall to \$40. Thus, if the duopolists individually pursue their own self-interest when deciding how much to produce, they produce a total quantity greater than the monopoly quantity, charge a price lower than the monopoly price, and earn total profit less than the monopoly profit.

Although the logic of self-interest increases the duopoly's output above the monopoly level, it does not push the duopolists all the way to the competitive allocation. Consider what happens when each duopolist produces 40 gallons. The price is \$40, and each duopolist makes a profit of \$1,600. In this case, Jack's self-interested logic leads to a different conclusion:

"Right now, my profit is \$1,600. Suppose I increase my production to 50 gallons. In this case, a total of 90 gallons of water would be sold, and the price would be \$30 a gallon. Then my profit would be only \$1,500. Rather than increasing production and driving down the price, I am better off keeping my production at 40 gallons."

The outcome in which Jack and Jill each produce 40 gallons looks like some sort of equilibrium. In fact, this outcome is called a Nash equilibrium. (It is named after Nobel Prize-winning mathematician and economic theorist John Nash, whose life was portrayed in the book and movie *A Beautiful Mind*.) A

Nash equilibrium (a situation in which economic actors interacting with one another each choose their best strategy given the strategies that all the other actors have chosen)

is a situation in which economic actors interacting with one another each choose their best strategy given the strategies that the others have chosen. In this case, given that Jill is producing 40 gallons, the best strategy for Jack is also to produce 40 gallons. Similarly, given that Jack is producing 40 gallons, the best strategy for Jill is also to produce 40 gallons. Once they reach this Nash equilibrium, neither Jack nor Jill has an incentive to make a different decision.

This example illustrates the tension between cooperation and self-interest. Oligopolists would be better off cooperating and reaching the monopoly outcome. Yet because they each pursue their own self-interest, they do not end up reaching the monopoly outcome and, thus, fail to maximize their joint profit. Each oligopolist is tempted to raise production and capture a larger share of the market. As each of them tries to do this, total production rises, and the price falls.

At the same time, self-interest does not drive the market all the way to the competitive outcome. Like monopolists, oligopolists are aware that increasing the amount they produce reduces the price of their product, which in turn affects profits. Therefore, they stop short of following the competitive firm's rule of producing up to the point where price equals marginal cost.

In summary, when firms in an oligopoly individually choose production to maximize profit, they produce a quantity of output greater than the level produced by monopoly and less than the level produced under perfect competition. The oligopoly price is less than the monopoly price but greater than the competitive price (which equals marginal cost).